PHYSICS

Higher Level

Set C | Paper 1A and 1B

Challenge mock examination

Time allowed: 120 minutes Maximum mark: 60

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• Paper 1A and Paper 1B share one uninterrupted examination time. Paper 1A has 40 multiple-choice questions; Paper 1B has two compulsory experimental/data questions worth 20 marks altogether.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

Thriving Scholars challenge practice | Current Physics syllabus | 2026

Paper 1A | Question paper

Paper 1A - Multiple choice

Select one answer, A, B, C or D, for each question. Use the answer grid or record your choices on the portal.

1	2	3	4	5	6	7	8	9	10

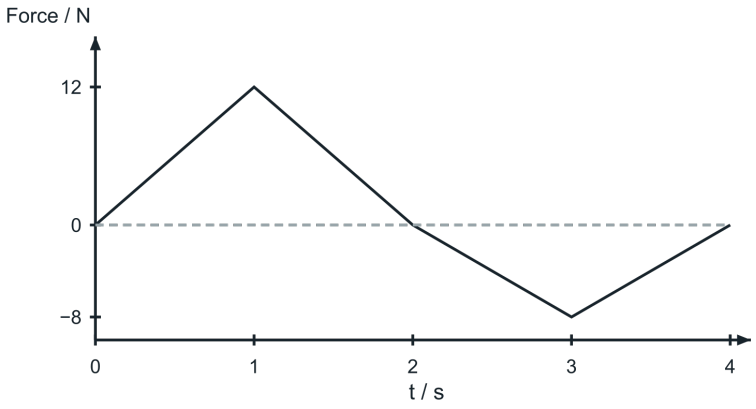
11	12	13	14	15	16	17	18	19	20

21	22	23	24	25	26	27	28	29	30

31	32	33	34	35	36	37	38	39	40

1A Question 1 [1 mark]

A $2.0\,\text{kg}$ trolley initially moves in the positive direction at $3.0\,\text{m s}^{-1}$. The graph shows the resultant force on it. What is its velocity at $t = 4.0\,\text{s}$?



A. $+5.0\,\text{m s}^{-1}$

B. $+7.0\,\text{m s}^{-1}$

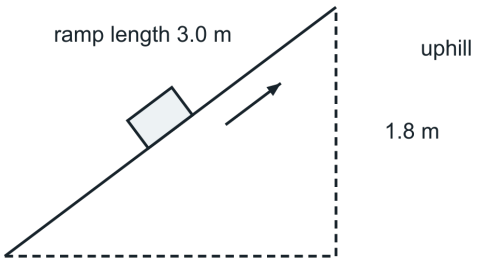
C. $+1.0\,\text{m s}^{-1}$

D. $+13\,\text{m s}^{-1}$

Paper 1A | Question paper

1A Question 2 [1 mark]

A $5.0\,\text{kg}$ crate slides from rest down the ramp shown, reaching $4.0\,\text{m}\,\text{s}^{-1}$ at the bottom. A constant resistive force opposes either direction of motion. A winch then returns the crate along the entire ramp at constant speed. What work does the winch do? Use $g = 10\,\text{m}\,\text{s}^{-2}$.



A. $90\,\text{J}$

B. $50\,\text{J}$

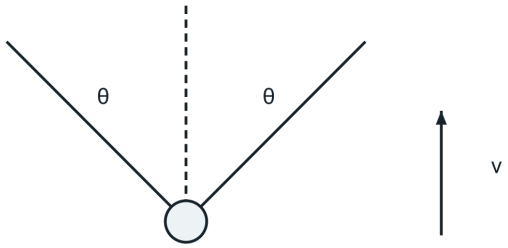
C. $140\,\text{J}$

D. $40\,\text{J}$

Paper 1A | Question paper

1A Question 3 [1 mark]

A load is raised vertically at constant speed by two symmetrical cables, each at angle θ to the vertical. The angle changes from 60° to 30°, and the vertical speed doubles. The load is unchanged and friction is negligible. What are the ratios of the new tension in one cable to its original tension, and of the new useful lifting power to its original value?



Two identical supporting cables

A. $1/\sqrt{3}$ and 4

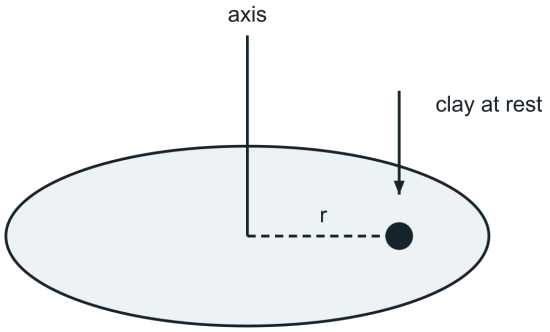
B. $\sqrt{3}$ and 1

C. $\sqrt{3}$ and 2

D. $1/\sqrt{3}$ and 2

1A Question 4 [1 mark]

A horizontal turntable has moment of inertia I and initial angular speed ω. Clay of mass m is dropped vertically and sticks at radius r, with $mr^2 = I$. In a separate identical trial it sticks at radius $r/2$. External torque about the axis is negligible. What is the ratio of rotational kinetic energy lost in the first trial to that lost in the second?



A. 4

B. $5/2$

C. $5/4$

D. 2

1A Question 5 [1 mark]

A uniform solid cylinder rolls without slipping from rest down a straight incline. Its moment of inertia is $I = \frac{1}{2}MR^2$. When its angular momentum about its centre has doubled, what is the factor by which its vertical drop has increased, and what fraction of its total kinetic energy is translational? The incline and g are constant.

A. 4 and $2/3$

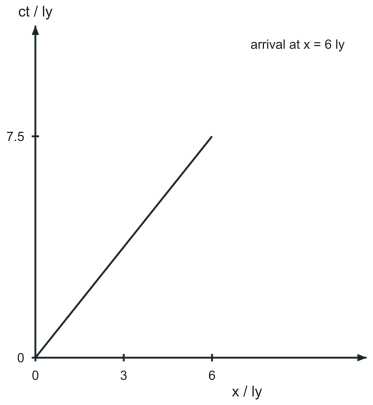
B. 2 and $2/3$

C. 4 and $1/2$

D. 2 and $1/3$

1A Question 6 [1 mark]

Two stations are $6.0\,\text{ly}$ apart in their common rest frame. A spacecraft travels between them at $0.80c$. What are the elapsed time on a single onboard clock and the station separation measured in the spacecraft frame, respectively?



A. $9.375\,\text{yr}$ and $3.6\,\text{ly}$

B. $7.5\,\text{yr}$ and $6.0\,\text{ly}$

C. $4.5\,\text{yr}$ and $6.0\,\text{ly}$

D. $4.5\,\text{yr}$ and $3.6\,\text{ly}$

1A Question 7 [1 mark]

A black surface of area A at temperature T radiates into surroundings at negligible temperature, with net power P. A second surface has area $2A$, temperature $2T$ and emissivity 0.25. It is surrounded by a large enclosure at temperature T. What is its net outward radiative power? Assume absorptivity equals emissivity.

A. $3.75P$

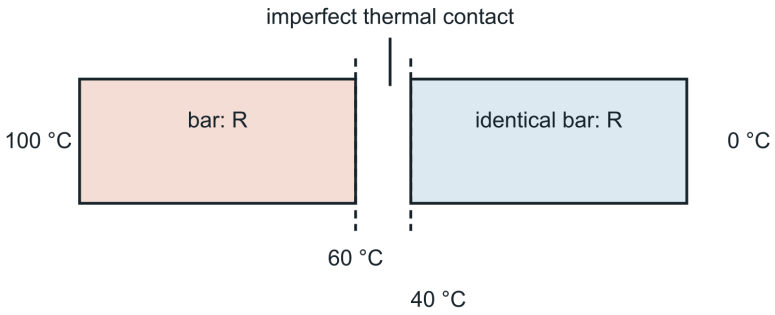
B. $8P$

C. $7.5P$

D. $0.5P$

1A Question 8 [1 mark]

Two identical bars conduct heat steadily between reservoirs at $100^\circ\mathrm{C}$ and $0^\circ\mathrm{C}$. Each bar has thermal resistance R. An imperfect contact separates them: its hot-side and cold-side temperatures are measured as $60^\circ\mathrm{C}$ and $40^\circ\mathrm{C}$. For any element, a supplied model is $\Delta T = \dot{Q} R_{\mathrm{th}}$, where $\dot{Q}$ is the heat-transfer rate. The contact resistance is then doubled while bar resistances and reservoir temperatures stay fixed. What are the new-to-old heat-rate ratio and the new hot-side contact temperature?



A. $2/3$ and $66.7^\circ\mathrm{C}$

B. $6/5$ and $33.3^\circ\mathrm{C}$

C. $5/6$ and $66.7^\circ\mathrm{C}$

D. $5/6$ and $60.0^\circ\mathrm{C}$

1A Question 9 [1 mark]

A planet is modelled with a single atmospheric layer that transmits incoming short-wave radiation and absorbs all outgoing long-wave radiation. The layer radiates equally upward and downward. The albedo increases from 0.20 to 0.36, while incident stellar flux and the atmospheric properties stay constant. What are the new-to-old surface-temperature ratio and the ratio of outgoing long-wave power at the top of the atmosphere to the upward long-wave power emitted by the surface?

A. 0.946 and 1

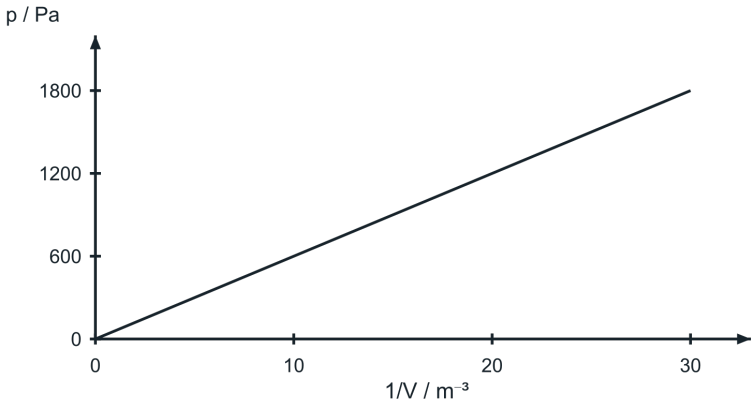
B. 0.800 and 1

C. 0.800 and $1/2$

D. 0.946 and $1/2$

1A Question 10 [1 mark]

The initial isotherm of a monatomic ideal gas is shown. The number of atoms is then doubled, and its absolute temperature is increased by a factor of 1.5. What is the final internal energy of the gas?



A. $540\ \text{J}$

B. $270\ \text{J}$

C. $90\ \text{J}$

D. $180\ \text{J}$

1A Question 11 [1 mark]

A monatomic ideal gas begins at A with pressure p_0 and volume V_0. It is compressed isothermally to B at volume $V_0/2$, expands adiabatically to C at volume V_0, and is heated at constant volume back to A. Use $pV^{5/3} = \text{constant}$ for an adiabat. What are the sign of the net work done by the gas and the ratio T_C/T_A?

A. Positive; $2^{2/3}$

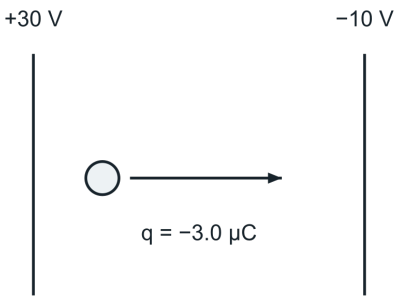
B. Negative; $2^{-2/3}$

C. Positive; $2^{-2/3}$

D. Negative; $2^{2/3}$

1A Question 12 [1 mark]

A particle of charge $-3.0\,\mu\text{C}$ moves from a point at $+30\,\text{V}$ to a point at $-10\,\text{V}$. Its initial kinetic energy is $200\,\mu\text{J}$, and only the electrostatic force does work. What is its final kinetic energy?



Only the electrostatic force does work

A. $120\,\mu\text{J}$

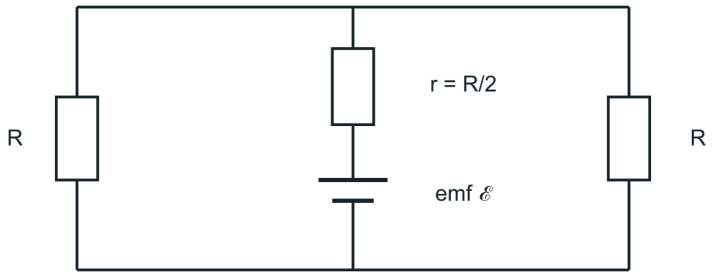
B. $320\,\mu\text{J}$

C. 0

D. $80\,\mu\text{J}$

1A Question 13 [1 mark]

Two resistors, each of resistance R, are in parallel across a cell of emf $\mathcal{E}$ and internal resistance $R/2$. One resistor is disconnected. What is the ratio of the total power dissipated in the remaining external circuit to the original total external power?



One external branch is disconnected later

A. 2

B. $8/9$

C. $4/9$

D. $1/2$

1A Question 14 [1 mark]

An oscillator has displacement $x = 0.060\cos(4t + \pi/3)$, with SI units. At an instant when $x = +0.030\,\text{m}$ and the oscillator is moving in the positive direction, what are its velocity and acceleration?

A. $+0.120\,\text{m s}^{-1}$ and $+0.480\,\text{m s}^{-2}$

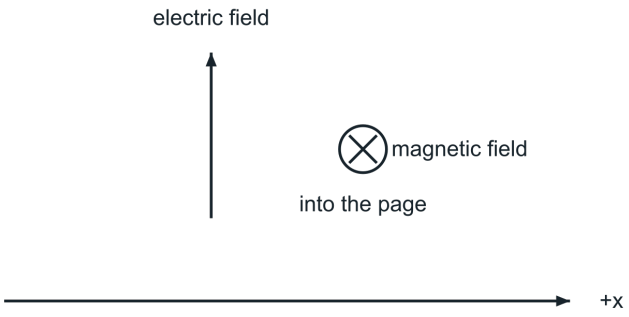
B. $-0.208\,\text{m s}^{-1}$ and $-0.480\,\text{m s}^{-2}$

C. $+0.208\,\text{m s}^{-1}$ and $-0.480\,\text{m s}^{-2}$

D. $+0.208\,\text{m s}^{-1}$ and $+0.480\,\text{m s}^{-2}$

1A Question 15 [1 mark]

At one instant, an electromagnetic wave has electric field upwards and magnetic field into the page, as shown. What are the magnetic-field direction half a period later and the propagation direction of the wave? Take right as positive x.



A. Out of the page; to the left

B. Into the page; to the left

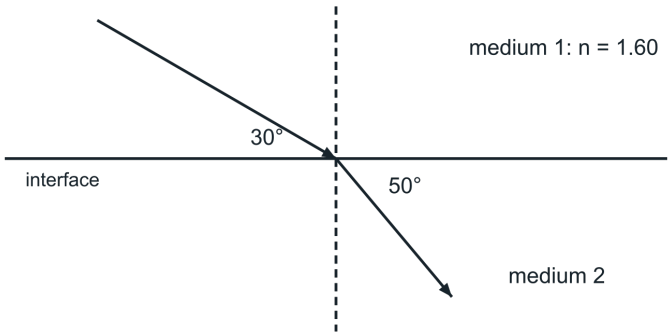
C. Into the page; to the right

D. Out of the page; to the right

Paper 1A | Question paper

1A Question 16 [1 mark]

A ray travels from medium 1 to medium 2. The labelled $30°$ and $50°$ are angles to the interface, not to the normal. The refractive index of medium 1 is 1.60. What is the critical angle, measured from the normal, for light travelling from medium 2 to medium 1?



A. $30.0°$

B. $42.1°$

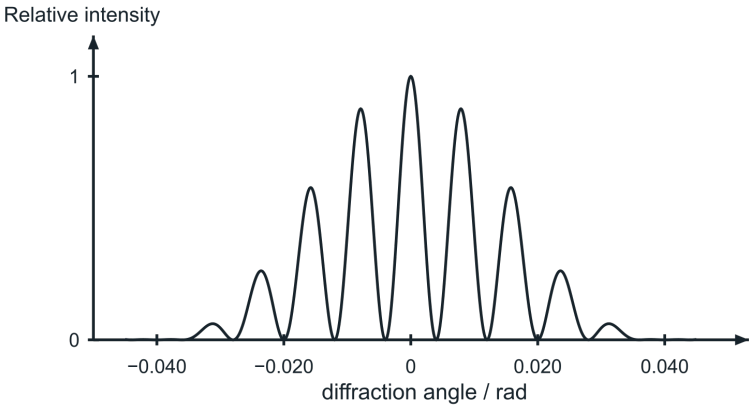
C. $47.9°$

D. $60.0°$

Paper 1A | Question paper

1A Question 17 [1 mark]

A double slit produces interference fringes with angular separation $0.0080\,\text{rad}$. The first single-slit envelope minima are at $\pm 0.040\,\text{rad}$. Use the small-angle approximation. How many integer-order constructive-interference directions, including the central direction, lie strictly between those first envelope minima?



A. 11

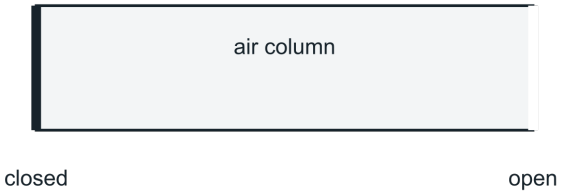
B. 5

C. 9

D. 10

1A Question 18 [1 mark]

Two consecutive resonances of an air column closed at one end occur at $210\ \text{Hz}$ and $350\ \text{Hz}$. The speed of sound is $350\ \text{m s}^{-1}$, and end corrections are negligible. What are the pipe length and the next resonance frequency above $350\ \text{Hz}$?



Consecutive resonances: 210 Hz, 350 Hz

A. $0.625\ \text{m}$ and $490\ \text{Hz}$

B. $0.417\ \text{m}$ and $630\ \text{Hz}$

C. $1.25\ \text{m}$ and $490\ \text{Hz}$

D. $1.25\ \text{m}$ and $420\ \text{Hz}$

1A Question 19 [1 mark]

A stellar spectral line has laboratory wavelength $500.00\ \text{nm}$. A spectrometer records the line at $500.35\ \text{nm}$. A stationary calibration source shows that every wavelength reading is $0.10\ \text{nm}$ too large. Using $c = 3.00 \times 10^{8}\ \text{m s}^{-1}$ and the low-speed Doppler approximation, what is the star's radial motion?

A. Approaching at $150\ \text{km s}^{-1}$

B. Receding at $210\ \text{km s}^{-1}$

C. Approaching at $210\ \text{km s}^{-1}$

D. Receding at $150\ \text{km s}^{-1}$

1A Question 20 [1 mark]

A horn has frequency $500\,\text{Hz}$ and the speed of sound in still air is $330\,\text{m}\,\text{s}^{-1}$. In experiment I the horn approaches a stationary observer at $66\,\text{m}\,\text{s}^{-1}$. In a separate experiment II the horn is stationary and the observer approaches it at $66\,\text{m}\,\text{s}^{-1}$. What is the received frequency in experiment I minus that in experiment II?

A. 0

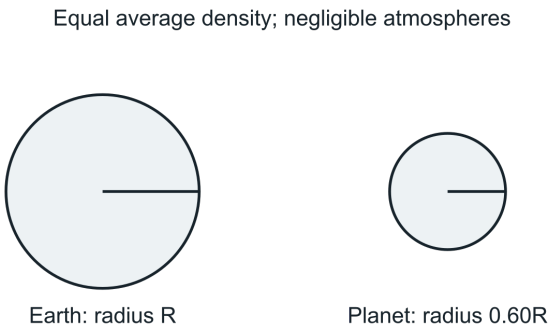
B. $25\,\text{Hz}$

C. $125\,\text{Hz}$

D. $100\,\text{Hz}$

1A Question 21 [1 mark]

A spherical planet has the same average density as Earth but radius 0.60 times Earth's radius. Atmospheric effects and rotation are negligible. What are the ratios of its surface gravitational field strength to Earth's and of a very low circular-orbit period to the corresponding period about Earth?



A. 0.36 and 0.60

B. 0.60 and $\sqrt{0.60}$

C. $1/0.60$ and 1

D. 0.60 and 1

1A Question 22 [1 mark]

Two isolated point masses m and $9m$ are separated by $4r$. A point Z between them is distance r from m, where the resultant gravitational field is zero. A small test mass μ is taken slowly from Z to infinity. What work is done by the external agent?



Objects isolated; potential is zero at infinity

A. $-4Gm\mu/r$

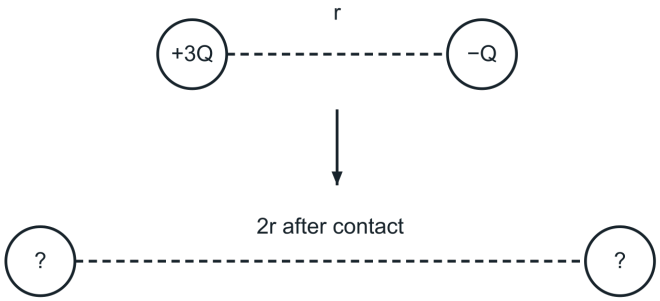
B. $+2Gm\mu/r$

C. 0

D. $+4Gm\mu/r$

1A Question 23 [1 mark]

Two identical conducting spheres initially have charges $+3Q$ and $-Q$, and their separation is r. The force magnitude is F. The spheres touch, are separated again, and are placed distance $2r$ apart. Their radii are negligible compared with either separation. What is the new force?



A. $F/12$, attractive

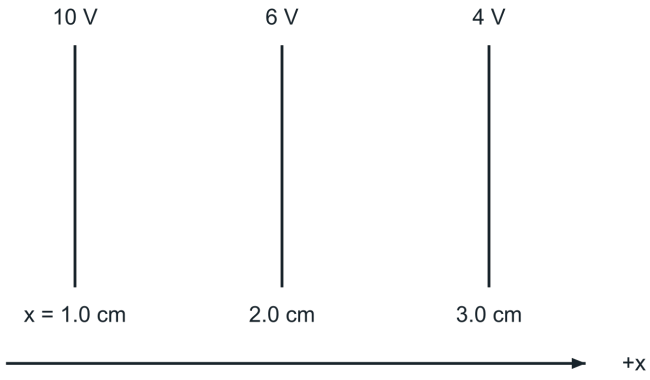
B. $3F/4$, repulsive

C. $F/3$, repulsive

D. $F/12$, repulsive

1A Question 24 [1 mark]

The diagram shows three equipotential surfaces in a region where the field is parallel to the x axis. A charge $+2.0\,\text{nC}$ moves from $x = 1.0\,\text{cm}$ to $x = 3.0\,\text{cm}$. What are the direction of the mean field in both intervals, the ratio of its mean magnitude in the first interval to that in the second, and the work done by the field on the charge?



A. $+x;\ 1/2;\ +12\,\text{nJ}$

B. $+x;\ 2;\ +12\,\text{nJ}$

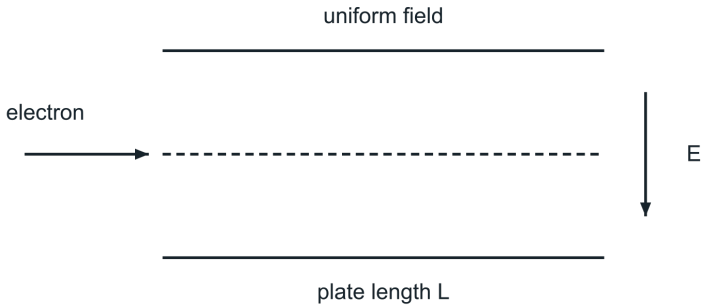
C. $-x;\ 2;\ +12\,\text{nJ}$

D. $+x;\ 2;\ -12\,\text{nJ}$

Paper 1A | Question paper

1A Question 25 [1 mark]

An electron is accelerated from rest through a potential difference V_a, then enters midway between parallel plates with horizontal speed v. The uniform plate field is perpendicular to v, and the electron exits without hitting a plate. The accelerating potential is increased to $4V_a$ and the plate length is doubled. The plate field remains unchanged, and the electron still exits. How do its exit deflection and the tangent of its exit angle compare with the originals?



A. Deflection quartered; tangent halved

B. Deflection unchanged; tangent halved

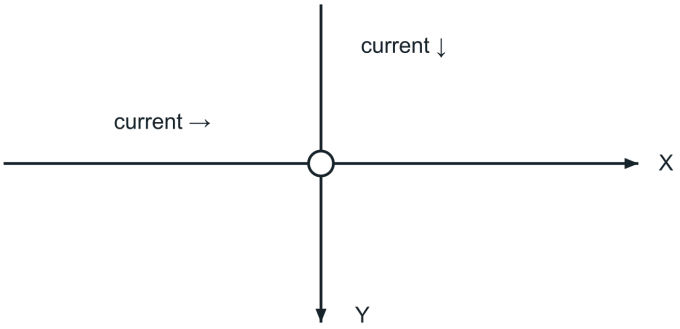
C. Deflection unchanged; tangent unchanged

D. Deflection halved; tangent halved

Paper 1A | Question paper

1A Question 26 [1 mark]

A long vertical straight wire Y carries current downwards. A finite horizontal wire X carries current to the right and is centred on Y. Both are in the plane of the page; a tiny insulated crossing can be neglected. What are the resultant magnetic force on X and its torque about the crossing, viewed as drawn?



Tiny insulated crossing; ignore its size

A. Zero force; anticlockwise torque

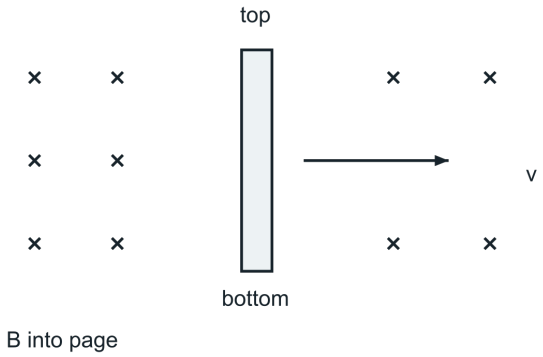
B. Zero force; clockwise torque

C. Downward force; zero torque

D. Upward force; zero torque

1A Question 27 [1 mark]

A vertical rod moves rightwards through a uniform magnetic field directed into the page. In open-circuit steady state, the magnitude of the potential difference between its ends is $\mathcal{E}$. Its speed is then tripled and its length halved, with the field unchanged. Which end is at higher potential, and what is the new potential-difference magnitude?



A. Bottom; $1.5\mathcal{E}$

B. Bottom; $6\mathcal{E}$

C. Top; $1.5\mathcal{E}$

D. Top; $6\mathcal{E}$

1A Question 28 [1 mark]

A rectangular generator coil has N turns, area A per turn and rotation period T in a constant uniform magnetic field. The normal to the coil rotates through all angles to the field. It is replaced by a coil with $2N$ turns and area $A/2$ per turn, rotating with period $T/2$. What are the ratios of peak induced emf and electrical frequency to their original values?

A. 2 and 2

B. 1 and 2

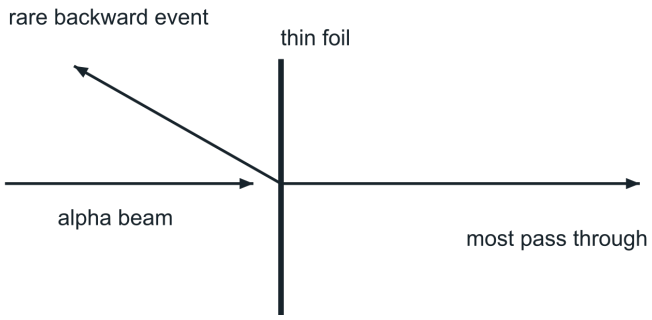
C. 2 and $1/2$

D. 4 and 2

Paper 1A | Question paper

1A Question 29 [1 mark]

Rare alpha particles reverse direction when a beam meets a thin metal foil. Test two possible single elastic encounters using a supplied head-on model: an alpha of mass M and initial velocity u meets a stationary target of mass m, leaving with velocity $v_\alpha = [(M - m)/(M + m)]u$. For an electron take $m = M/7000$; for a heavy nucleus take $m = 50M$. Which approximate velocity pair and conclusion follow?



A. Electron target: $+0.9997u$; nuclear target: $-0.961u$; only the heavy target can reverse the alpha

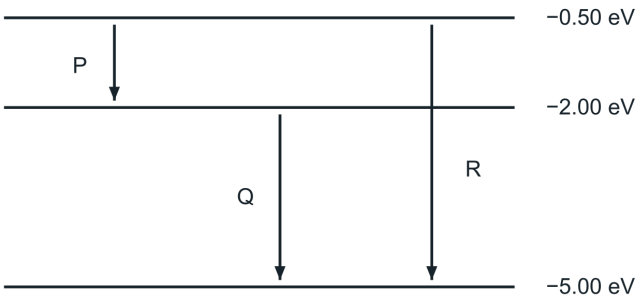
B. Electron target: $+0.9997u$; nuclear target: $+0.961u$; neither can reverse the alpha

C. Electron target: $-0.9997u$; nuclear target: $+0.961u$; only the electron can reverse the alpha

D. Electron target: $-0.9997u$; nuclear target: $-0.961u$; either can reverse the alpha

1A Question 30 [1 mark]

The atom has the energy levels and downward transitions shown. Take visible wavelengths as 400–$700\,\text{nm}$ and use $hc = 1240\,\text{eV nm}$. Which transition emits visible light, and which emits ultraviolet light?



A. Q visible; R ultraviolet

B. Q visible; P ultraviolet

C. P visible; Q ultraviolet

D. R visible; P ultraviolet

1A Question 31 [1 mark]

A photon of wavelength $\lambda_C = h/(m_e c)$ is scattered by a stationary electron. Its wavelength increases by $1.5\lambda_C$. Use $\Delta\lambda = \lambda_C(1 - \cos\theta)$ and neglect binding effects. What are the photon scattering angle and the fraction of the incident photon energy transferred to the electron?

A. 60° and 0.60

B. 60° and 0.40

C. 120° and 0.60

D. 120° and 1.50

1A Question 32 [1 mark]

Electrons start from rest and are accelerated through $900\,\text{V}$. They then pass through a retarding electric potential difference without stopping. Their de Broglie wavelength after retardation is 50% greater than immediately before it. Speeds are non-relativistic. What is the magnitude of the retarding potential difference?

A. $500\,\text{V}$

B. $400\,\text{V}$

C. $600\,\text{V}$

D. $300\,\text{V}$

1A Question 33 [1 mark]

A uranium nucleus $^{238}_{92}\mathrm{U}$ undergoes one alpha decay followed by two beta-minus decays. Gamma emissions, if any, can be ignored. How many protons and neutrons are in the final nucleus Z?



A. 92 protons and 142 neutrons

B. 94 protons and 140 neutrons

C. 92 protons and 146 neutrons

D. 90 protons and 144 neutrons

1A Question 34 [1 mark]

A $^{13}_{7}\mathrm{N}$ nucleus undergoes beta-plus decay. The available kinetic energy is $1.20\,\mathrm{MeV}$. In one event the positron receives $0.70\,\mathrm{MeV}$; daughter recoil can be neglected. What are the daughter proton number and the energy carried by the electron neutrino?

A. 6 and $0.50\,\mathrm{MeV}$

B. 6 and $1.90\,\mathrm{MeV}$

C. 8 and $0.50\,\mathrm{MeV}$

D. 8 and $1.20\,\mathrm{MeV}$

1A Question 35 [1 mark]

A nuclide has half-life $18\,\mathrm{h}$. What is the probability that a particular undecayed nucleus decays in the next $6.0\,\mathrm{h}$, and how does $\lambda\Delta t$ compare with that probability?

A. 0.333; $\lambda\Delta t$ is larger

B. 0.794; $\lambda\Delta t$ is smaller

C. 0.206; $\lambda\Delta t$ is larger

D. 0.231; $\lambda\Delta t$ is exactly equal

1A Question 36 [1 mark]

For a head-on elastic collision between a neutron and a stationary nucleus of mass Am_n, a supplied model gives neutron energy ratio $K_f/K_i = [(A-1)/(A+1)]^2$. A neutron makes two such collisions with nuclei of $A = 2$. Which remaining-energy fraction and material property favour effective moderation?

A. $1/81$; unstable nuclei with a high neutron-absorption probability

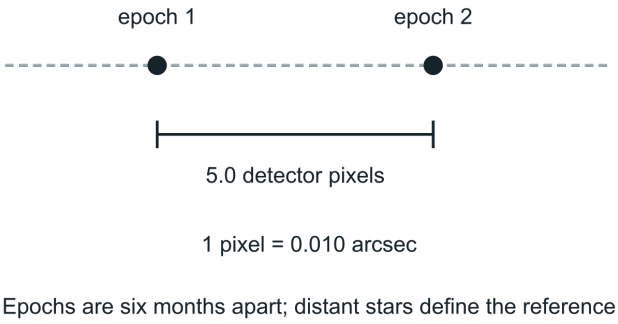
B. $1/81$; stable nuclei with a low neutron-absorption probability

C. $8/9$; stable nuclei with a low neutron-absorption probability

D. $1/9$; nuclei with a high neutron-absorption probability

1A Question 37 [1 mark]

A nearby star shifts by 5.0 detector pixels relative to distant background stars between observations made six months apart at opposite ends of Earth's orbit. The scale is 0.010 arc-second per pixel. Proper motion is negligible. What is the star's distance?



A. $100\,\text{pc}$

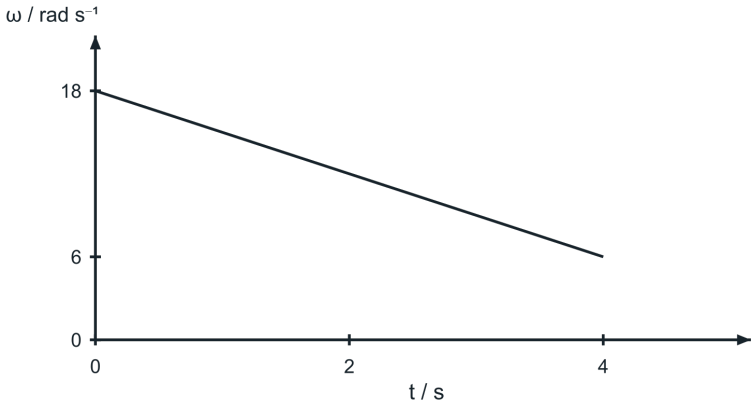
B. $20\,\text{pc}$

C. $200\,\text{pc}$

D. $40\,\text{pc}$

1A Question 38 [1 mark]

A disconnected flywheel slows under a constant bearing-friction torque. Its angular-speed record is shown. Friction dissipates $108\,\text{J}$ during the first $4.0\,\text{s}$. No other energy transfers occur. What are its angular momentum at $t = 0$ and its further angular displacement from $t = 4.0\,\text{s}$ until it stops?



A. $13.5\,\text{kg}\,\text{m}^2\,\text{s}^{-1}$ and $24\,\text{rad}$

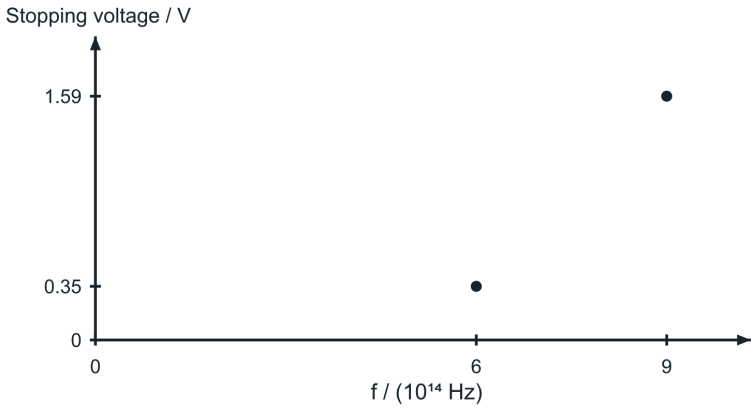
B. $27.0\,\text{kg}\,\text{m}^2\,\text{s}^{-1}$ and $12\,\text{rad}$

C. $108\,\text{kg}\,\text{m}^2\,\text{s}^{-1}$ and $6.0\,\text{rad}$

D. $13.5\,\text{kg}\,\text{m}^2\,\text{s}^{-1}$ and $6.0\,\text{rad}$

1A Question 39 [1 mark]

A photoelectric experiment gives stopping voltages $0.35\,\text{V}$ and $1.59\,\text{V}$ at frequencies $6.0 \times 10^{14}\,\text{Hz}$ and $9.0 \times 10^{14}\,\text{Hz}$. The same clean metal surface is used. What is its approximate threshold frequency, and what stopping voltage is expected if the light intensity is doubled at $6.0 \times 10^{14}\,\text{Hz}$?



A. $5.15 \times 10^{14}\,\text{Hz}$ and $0.35\,\text{V}$

B. $6.0 \times 10^{14}\,\text{Hz}$ and $0.70\,\text{V}$

C. $5.15 \times 10^{14}\,\text{Hz}$ and $0.70\,\text{V}$

D. $9.0 \times 10^{14}\,\text{Hz}$ and $0.35\,\text{V}$

1A Question 40 [1 mark]

An engine always achieves 75% of the Carnot efficiency between its reservoirs. It absorbs $1200\,\mathrm{J}$ per cycle. The cold reservoir stays at $27^\circ\mathrm{C}$, while the hot reservoir increases from $327^\circ\mathrm{C}$ to $477^\circ\mathrm{C}$. Use $T/\mathrm{K} = t/^\circ\mathrm{C} + 273$. By how much does the work per cycle increase?

A. $180\,\mathrm{J}$

B. $90\,\mathrm{J}$

C. $120\,\mathrm{J}$

D. $540\,\mathrm{J}$

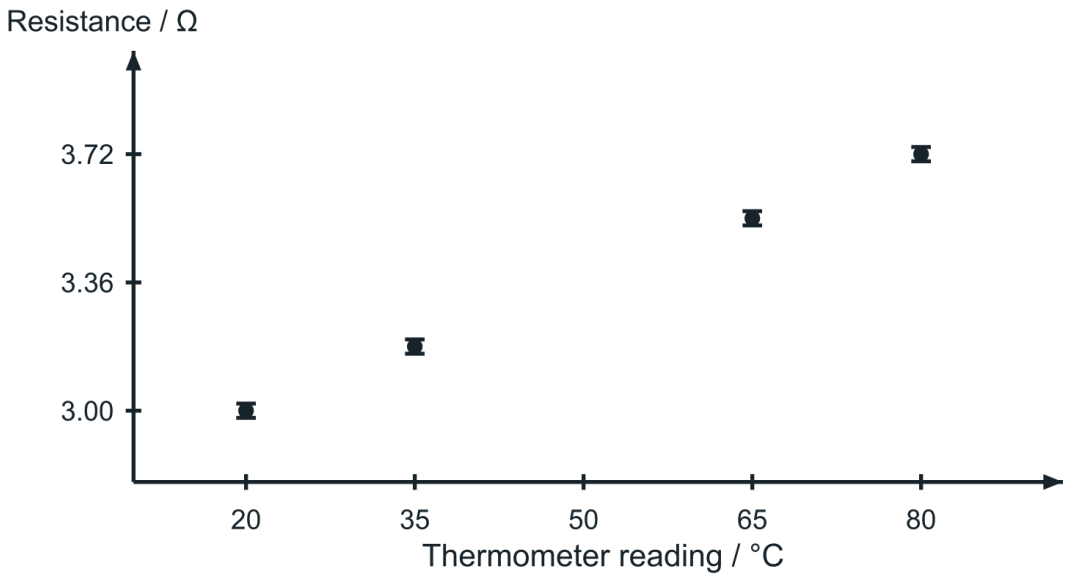
1B Question 1 | Question paper

Paper 1B - Experimental and data-based questions

Continue within the same examination time. Answer both questions. Use the data, graphs and uncertainty information provided.

1B Question 1 [10 marks]

A laboratory group measures the resistance of an insulated metal wire immersed in a slowly cooling, stirred water bath. The graph shows synthetic laboratory data. All resistance error bars are $\pm 0.020\ \Omega$; temperature uncertainty is negligible for parts (a)–(d). At the missing thermometer reading of $50.0\,^{\circ}\mathrm{C}$, the wire potential difference is $0.840\ \mathrm{V}$ and current is $0.250\ \mathrm{A}$. The wire length is $(2.00 \pm 0.01)\ \mathrm{m}$ and its diameter is $(0.400 \pm 0.004)\ \mathrm{mm}$; thermal expansion is negligible.



(a) Calculate the resistance at the missing data point. [1]

(b) Using the points at $20.0\,^{\circ}\mathrm{C}$ and $80.0\,^{\circ}\mathrm{C}$, determine the best gradient of resistance against thermometer reading and its uncertainty from the steepest and shallowest endpoint lines. [2]

(c) Calculate the resistivity at the $50.0\,^{\circ}\mathrm{C}$ thermometer reading. [2]

1B Question 1 | Question paper

(d) Use maximum fractional uncertainties to state the resistivity at this reading with its absolute uncertainty and SI unit. [2]

(e) An ice–water check subsequently gives a thermometer reading of $-1.5\,^{\circ}\mathrm{C}$. Identify the error type and describe how correcting it changes the horizontal positions of all points and the gradient. [2]

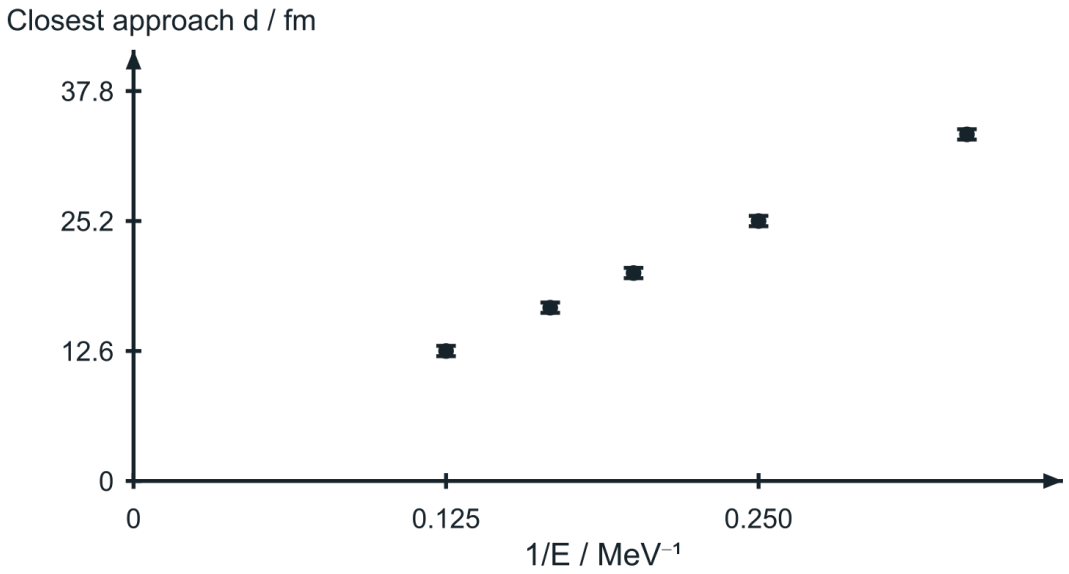
(f) State one reason for stirring the bath while readings are taken. [1]

1B Question 2 | Question paper

1B Question 2 [10 marks]

A research-training simulation produces synthetic data for head-on alpha scattering from heavy stationary nuclei. Outside the nucleus, the purely electrostatic model predicts $d = K/E$ and $K = 2Z(1.44\,\text{MeV fm})$, where E is incident alpha kinetic energy, d is closest approach, and Z is target proton number. Binding and target recoil are neglected. Each d has uncertainty $\pm 0.5\,\text{fm}$; energy uncertainty is negligible. The reciprocal-energy graph represents these data.

E/MeV	3.00	4.00	5.00	6.00	8.00
d/fm	33.6	25.2	20.16	16.8	12.6



(a) Explain why the model predicts a straight line through the origin on these axes. [1]

(b) Use the 3.00 and $8.00\,\text{MeV}$ endpoints to calculate K and its uncertainty from the steepest and shallowest endpoint lines. [2]

(c) Calculate the target proton number and its uncertainty implied by part (b). [2]

1B Question 2 | Question paper

(d) The target nucleon number is 80 and its radius is modelled by $R = 1.20A^{1/3}\,\text{fm}$. Estimate the incident energy at which electrostatic closest approach would equal this radius. [2]

(e) At $12.0\,\text{MeV}$ a new simulation gives $d = (6.0 \pm 0.5)\,\text{fm}$. Test its consistency with the electrostatic prediction using your uncertainty in K. State one justified limitation of interpreting the discrepancy as proof of a particular nuclear mechanism. [3]

End of question paper

PHYSICS

Higher Level

Set C | Paper 1A and 1B

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 1A - Answer key and explanations

1	2	3	4	5	6	7	8	9	10
A	C	D	B	A	D	C	C	D	B

11	12	13	14	15	16	17	18	19	20
B	D	B	C	A	C	C	C	D	B

21	22	23	24	25	26	27	28	29	30
D	D	D	B	B	B	C	A	A	A

31	32	33	34	35	36	37	38	39	40
C	A	A	A	C	B	D	D	A	B

1A Question 1 [1 mark]

Correct answer: A

The positive impulse is $\frac{1}{2}(2)(12) = 12\,\text{N s}$ and the negative impulse is $-\frac{1}{2}(2)(8) = -8\,\text{N s}$. Thus $p_f = 2(3) + 4 = 10\,\text{kg m s}^{-1}$, giving $v_f = +5.0\,\text{m s}^{-1}$. Treating both areas as positive or forgetting initial momentum gives the wrong result. Therefore choose A.

1A Question 2 [1 mark]

Correct answer: C

Descent releases $mgh = 90\,\text{J}$ and leaves $\frac{1}{2}mv^2 = 40\,\text{J}$. Hence $50\,\text{J}$ is dissipated over the ramp. Ascent at constant speed requires $90 + 50 = 140\,\text{J}$: friction still opposes motion and does not return its dissipated energy. Therefore choose C.

1A Question 3 [1 mark]

Correct answer: D

Vertical equilibrium gives $2T\cos\theta = mg$, so $T_{30}/T_{60} = \cos 60^\circ / \cos 30^\circ = 1/\sqrt{3}$. Useful power is mgv, so doubling vertical speed doubles it. The cable tension alone is not the net upward force. Therefore choose D.

1A Question 4 [1 mark]

Correct answer: B

Conservation of angular momentum gives $\omega_f = I\omega/(I + mr^2)$ and $\Delta K/K_i = mr^2/(I + mr^2)$. The losses are $K_i/2$ and $K_i/5$, respectively. Their ratio is $5/2$, not 4: the final angular speed also changes with radius. Therefore choose B.

1A Question 5 [1 mark]

Correct answer: A

As $L = I\omega$, doubling L doubles ω and, through $v = R\omega$, doubles v. Both kinetic-energy terms therefore quadruple, so the vertical drop quadruples by energy conservation. $K = \frac{1}{2}Mv^2 + \frac{1}{4}Mv^2$, giving $K_{\text{trans}}/K = 2/3$. Therefore choose A.

1A Question 6 [1 mark]

Correct answer: D

The station-frame duration is $6/0.8 = 7.5\,\text{yr}$. With $\gamma = 5/3$, the onboard proper time is $7.5/\gamma = 4.5\,\text{yr}$ and the simultaneous spacecraft-frame separation is $6/\gamma = 3.6\,\text{ly}$. The station separation is the proper length; the onboard elapsed interval is the proper time. Therefore choose D.

1A Question 7 [1 mark]

Correct answer: C

The first power is $P = \sigma A T^4$. The second net power is $\epsilon(2A)\sigma[(2T)^4 - T^4] = 0.25 \times 2 \times 15P = 7.5P$. The $8P$ distractor neglects absorbed enclosure radiation; using the temperature difference to the fourth power is also incorrect. Therefore choose C.

1A Question 8 [1 mark]

Correct answer: C

Initially each bar has a $40\,\text{K}$ drop, while the contact has $20\,\text{K}$, so its resistance is $R/2$. Total resistance is $2.5R$. Doubling only the contact makes its resistance R and the total $3R$, so $\dot{Q}_{\text{new}}/\dot{Q}_{\text{old}} = 2.5/3 = 5/6$. The three equal new resistances divide the $100\,\text{K}$ drop equally: the hot-side contact face is at $100 - 100/3 = 66.7^\circ\text{C}$. The old contact temperatures do not remain fixed. Therefore choose C.

1A Question 9 [1 mark]

Correct answer: D

At layer equilibrium, two equal layer emissions balance the absorbed surface emission, so outward top-of-atmosphere long-wave power is half the surface emission. In both states surface T^4 is proportional to $(1-a)$, including the same greenhouse factor. Thus $T_2/T_1 = (0.64/0.80)^{1/4} = 0.946$. A temperature ratio is not the absorbed-power ratio. Therefore choose D.

1A Question 10 [1 mark]

Correct answer: B

The gradient of p against $1/V$ is $nRT = 1200/20 = 60\,\text{J}$. Initially $U = 3nRT/2 = 90\,\text{J}$. The product nT triples, so the final energy is $270\,\text{J}$. A pressure-reciprocal-volume gradient has units $\text{Pa}\,\text{m}^3 = \text{J}$, not pressure. Therefore choose B.

1A Question 11 [1 mark]

Correct answer: B

At B, $p_B = 2p_0$ and $T_B = T_A$. The adiabat gives $p_C = 2p_0(1/2)^{5/3} = 2^{-2/3}p_0$, hence $T_C/T_A = 2^{-2/3}$ at equal volumes. Compression follows the higher-pressure isotherm and expansion the lower-pressure adiabat, so the negative compression work exceeds the positive expansion work: net work by the gas is negative. Therefore choose B.

1A Question 12 [1 mark]

Correct answer: D

$\Delta U = q\Delta V = (-3.0)(-40) = +120\,\mu\text{J}$. The field therefore does $-120\,\mu\text{J}$ of work and $K_f = 200 - 120 = 80\,\mu\text{J}$. A negative charge has increasing potential energy on moving to a lower electric potential. Therefore choose D.

1A Question 13 [1 mark]

Correct answer: B

Initially $R_{\text{ext}} = R/2$, the terminal voltage is $\mathcal{E}/2$ and $P_i = \mathcal{E}^2/(2R)$. Afterwards $R_{\text{ext}} = R$, the terminal voltage becomes $2\mathcal{E}/3$ and $P_f = 4\mathcal{E}^2/(9R)$. Thus $P_f/P_i = 8/9$. The terminal voltage is not constant for a cell with internal resistance. Therefore choose B.

1A Question 14 [1 mark]

Correct answer: C

$|v| = \omega\sqrt{A^2 - x^2} = 4\sqrt{0.060^2 - 0.030^2} = 0.208\,\text{m s}^{-1}$. The stated motion selects the positive sign. Acceleration is $a = -\omega^2 x = -16(0.030) = -0.480\,\text{m s}^{-2}$, always towards equilibrium. Therefore choose C.

1A Question 15 [1 mark]

Correct answer: A

The electric and magnetic fields of the wave are in phase and both reverse after half a period. Thus the later magnetic field is out of the page. At either time, the mutually perpendicular field directions imply propagation to the left: initially upwards crossed with into the page points left. Reversing both fields leaves the propagation direction unchanged. Therefore choose A.

1A Question 16 [1 mark]

Correct answer: C

The normal angles are 60° and 40°. Snell's law gives $n_2 = 1.60 \sin 60^\circ / \sin 40^\circ = 2.16$. On reversing towards the lower-index medium, $\sin c = n_1/n_2 = \sin 40^\circ / \sin 60^\circ$, so $c = 47.9^\circ$. The complement is not the requested critical angle. Therefore choose C.

1A Question 17 [1 mark]

Correct answer: C

Integer constructive-interference orders occur at $\theta_m = m(0.0080)$. The envelope boundaries coincide with $m = \pm 5$, where the intensity is zero, so these maxima are missing. The allowed indices strictly inside are $-4, -3, \ldots, 3, 4$, giving nine allowed integer orders including the central direction. The exact peaks of the combined envelope need not lie at these integer-order angles. Therefore choose C.

1A Question 18 [1 mark]

Correct answer: C

A closed–open pipe has odd multiples of f_0, so consecutive resonances differ by $2f_0 = 140\,\text{Hz}$. Hence $f_0 = 70\,\text{Hz}$ and $L = v/(4f_0) = 1.25\,\text{m}$. The given resonances are $3f_0$ and $5f_0$; the next is $7f_0 = 490\,\text{Hz}$. Therefore choose C.

Paper 1A | Worked solutions

1A Question 19 [1 mark]

Correct answer: D

The corrected stellar wavelength is $500.25\,\text{nm}$, so $\Delta\lambda/\lambda = 0.25/500 = 5.0 \times 10^{-4}$. Hence $v = c\Delta\lambda/\lambda = 1.50 \times 10^{5}\,\text{m}\,\text{s}^{-1}$. The corrected wavelength is still longer than the laboratory wavelength, so the star recedes. Using the uncorrected shift gives $210\,\text{km}\,\text{s}^{-1}$. Therefore choose D.

1A Question 20 [1 mark]

Correct answer: B

A moving source compresses the wavelength: $f_I = fv/(v-u) = 500(330)/(330-66) = 625\,\text{Hz}$. A moving observer meets unchanged wavefront spacing at a different rate: $f_{II} = f(v+u)/v = 500(330+66)/330 = 600\,\text{Hz}$. The difference is $25\,\text{Hz}$. Sound's stationary medium makes these experiments inequivalent; no experiment has both source and observer moving. Therefore choose B.

1A Question 21 [1 mark]

Correct answer: D

At fixed density, $M \propto R^3$, so $g = GM/R^2 \propto R$ and the field ratio is 0.60. From circular motion $T = 2\pi\sqrt{R^3/(GM)}$, which is independent of radius at fixed density. Thus both very low orbit periods are equal. Therefore choose D.

1A Question 22 [1 mark]

Correct answer: D

The field cancellation follows from $Gm/r^2 = 9Gm/(3r)^2$, but potentials add as scalars: $V_Z = -Gm/r - 9Gm/(3r) = -4Gm/r$. Slow removal requires work $\Delta U = \mu(0 - V_Z) = +4Gm\mu/r$. A zero local gradient does not make the potential zero. Therefore choose D.

1A Question 23 [1 mark]

Correct answer: D

Total charge $+2Q$ divides equally, leaving $+Q$ on each sphere. Initially $F = 3kQ^2/r^2$. Finally $F_f = kQ^2/(2r)^2 = kQ^2/(4r^2) = F/12$, and equal positive charges repel. Both charge redistribution and the squared-distance change must be included. Therefore choose D.

1A Question 24 [1 mark]

Correct answer: B

The field points towards decreasing potential, so it is along $+x$. Equal interval widths contain potential drops of $4\,\text{V}$ and $2\,\text{V}$, giving mean field magnitudes in ratio $2:1$. The field work is $q(V_i - V_f) = 2.0(10-4) = +12\,\text{nJ}$. Therefore choose B.

1A Question 25 [1 mark]

Correct answer: B

Acceleration through V_a gives $v \propto \sqrt{V_a}$, so the new horizontal speed is $2v$. Doubling the plate length leaves the transit time L/v unchanged; hence the transverse velocity and deflection are unchanged. Since $\tan\theta = v_y/v_x$, doubling v_x while retaining v_y halves the tangent. Therefore choose B.

1A Question 26 [1 mark]

Correct answer: B

For downward current in Y, its field is out of the page on the right and into the page on the left. With current in X to the right, the right half experiences downward force and the left half upward force. Equal and opposite force contributions cancel, but both halves produce clockwise torque about the crossing. Therefore choose B.

1A Question 27 [1 mark]

Correct answer: C

The force on positive charges is $q\mathbf{v} \times \mathbf{B}$, upwards here, so the top becomes positive and is at higher potential. Charge separation creates an opposing electric field. The open-circuit emf is $B\ell v$, and the new magnitude is $B(\ell/2)(3v) = 1.5\mathcal{E}$. Therefore choose C.

1A Question 28 [1 mark]

Correct answer: A

The peak induced emf is $NBA\omega$. The factors from turns, area and angular speed are 2, $1/2$ and 2, giving an overall factor of 2. A complete revolution produces one complete sinusoidal cycle, so halving rotation period doubles the electrical frequency. Therefore choose A.

1A Question 29 [1 mark]

Correct answer: A

For the electron target, $v_\alpha/u = (7000-1)/(7000+1) = +0.999714$: the alpha remains forward-moving. For the heavy target, $v_\alpha/u = (1-50)/(1+50) = -49/51 = -0.960784$, so it reverses. The single-encounter model therefore rules out the much lighter electrons as the cause of strong alpha reversal; a sufficiently massive compact scattering region is required. The equation is supplied, so no simultaneous collision-energy/momentum derivation is required. Therefore choose A.

1A Question 30 [1 mark]

Correct answer: A

The photon energies are $E_P = 1.50\,\text{eV}$, $E_Q = 3.00\,\text{eV}$ and $E_R = 4.50\,\text{eV}$. Their wavelengths are approximately 827, 413 and $276\,\text{nm}$. Thus P is infrared, Q visible and R ultraviolet. A two-photon cascade conserves the same total energy as R but does not emit a single R-frequency photon. Therefore choose A.

1A Question 31 [1 mark]

Correct answer: C

$1-\cos\theta = 1.5$ gives $\cos\theta = -0.5$, so $\theta = 120^\circ$. The scattered wavelength is $2.5\lambda_C$, making the photon energy $1/2.5 = 0.40$ of its initial value. Energy conservation transfers the remaining fraction 0.60 to electron kinetic energy. The wavelength increase is not an energy fraction. Therefore choose C.

1A Question 32 [1 mark]

Correct answer: A

Non-relativistically $\lambda = h/\sqrt{2mK}$, so $K_f/K_i = (\lambda_i/\lambda_f)^2 = (1/1.5)^2 = 4/9$. Initially $K_i = 900\,\text{eV}$; afterwards it is $400\,\text{eV}$. The electric field therefore removes $500\,\text{eV}$ per electron, corresponding to a retarding potential magnitude of $500\,\text{V}$. The wavelength increase is not the fractional energy loss. Therefore choose A.

1A Question 33 [1 mark]

Correct answer: A

Alpha decay changes (A, Z) from $(238, 92)$ to $(234, 90)$. Each beta-minus decay converts a neutron to a proton, increasing Z by one without changing A. The final values are $A = 234$, $Z = 92$, and $N = A - Z = 142$. Gamma emission changes neither. Therefore choose A.

1A Question 34 [1 mark]

Correct answer: A

In beta-plus decay a proton becomes a neutron while a positron and an electron neutrino are emitted, so the daughter has Z=6 and unchanged A=13. Energy conservation leaves $1.20 - 0.70 = 0.50\,\text{MeV}$ for the neutrino when recoil is neglected. The variable sharing is why beta energies are continuous. Therefore choose A.

1A Question 35 [1 mark]

Correct answer: C

The survival probability is $e^{-\lambda \Delta t} = 2^{-6/18}$, so the decay probability is $1 - 2^{-1/3} = 0.206$. Here $\lambda \Delta t = (\ln 2)/3 = 0.231$, which overestimates the probability. The linear expression is only a small-interval approximation; surviving population decreases throughout the interval. Therefore choose C.

1A Question 36 [1 mark]

Correct answer: B

One collision leaves $[(2-1)/(2+1)]^2 = 1/9$ of the neutron energy; two leave $(1/9)^2 = 1/81$. Moderation should slow neutrons while retaining them for the chain reaction, so stability and a low absorption probability are desirable. Absorbing neutrons is primarily the role of control materials, not the required moderation mechanism. Therefore choose B.

1A Question 37 [1 mark]

Correct answer: D

The full six-month displacement is $5.0(0.010) = 0.050$ arc-second, equal to $2p$, where p is the parallax angle for a baseline of one astronomical unit. Thus $p = 0.025$ arc-second and $d = 1/p = 40\,\text{pc}$. Inverting the full displacement would give the erroneous 20 pc. Therefore choose D.

1A Question 38 [1 mark]

Correct answer: D

The graph gives $\alpha = (6 - 18)/4 = -3.0\,\text{rad}\,\text{s}^{-2}$. The energy loss is $108 = \frac{1}{2} I (18^2 - 6^2)$, so $I = 0.750\,\text{kg}\,\text{m}^2$ and $L_0 = I\omega_0 = 13.5\,\text{kg}\,\text{m}^2\,\text{s}^{-1}$. From $0 = 6^2 + 2(-3)\Delta\theta$, the remaining angle is $6.0\,\text{rad}$. The speed-time gradient alone does not give the moment of inertia; the measured dissipated energy is essential. Therefore choose D.

1A Question 39 [1 mark]

Correct answer: A

From $V_s = (h/e)(f - f_0)$, the gradient is $(1.59 - 0.35)/(3.0 \times 10^{14}) = 4.13 \times 10^{-15}\,\text{V s}$. Hence $f_0 = 6.0 \times 10^{14} - 0.35/(4.13 \times 10^{-15}) = 5.15 \times 10^{14}\,\text{Hz}$. Intensity changes the number of photons, not individual photon energy, so the stopping voltage remains $0.35\,\text{V}$. Therefore choose A.

1A Question 40 [1 mark]

Correct answer: B

The kelvin temperatures are $T_C = 300$, $T_{H1} = 600$ and $T_{H2} = 750\,\text{K}$. Actual efficiencies are $0.75(1 - 300/600) = 0.375$ and $0.75(1 - 300/750) = 0.450$. Work rises from 450 to $540\,\text{J}$, an increase of $90\,\text{J}$. The increase in ideal Carnot work would be $120\,\text{J}$, not the actual increase. Therefore choose B.

Paper 1B - Worked solutions

1B Question 1 [10 marks]

(a) [1 mark]

A1 $R = V/I = 0.840/0.250 = 3.36\,\Omega$.

(b) [2 marks]

A1 $g = (3.72 - 3.00)/(80 - 20) = 0.0120\,\Omega\,^{\circ}\mathrm{C}^{-1}$.

A1 $g_{\max} = (3.74 - 2.98)/60 = 0.01267$ and $g_{\min} = (3.70 - 3.02)/60 = 0.01133\,\Omega\,^{\circ}\mathrm{C}^{-1}$; half the range gives $\Delta g = 0.00067\,\Omega\,^{\circ}\mathrm{C}^{-1}$. Accept equivalent clearly drawn limiting lines.

(c) [2 marks]

M1 $A = \pi d^2/4 = \pi(0.400 \times 10^{-3})^2/4 = 1.2566 \times 10^{-7}\,\mathrm{m}^2$, and $\rho = RA/L$.

A1 $\rho = 3.36(1.2566 \times 10^{-7})/2.00 = 2.11 \times 10^{-7}\,\Omega\,\mathrm{m}$.

(d) [2 marks]

M1 $\Delta\rho/\rho = \Delta R/R + 2\Delta d/d + \Delta L/L = 0.020/3.36 + 2(0.004/0.400) + 0.01/2.00 = 0.03095$.

A1 $\Delta\rho = 6.53 \times 10^{-9}\,\Omega\,\mathrm{m}$, so quote $(2.11 \pm 0.07) \times 10^{-7}\,\Omega\,\mathrm{m}$; a consistently rounded uncertainty of 0.065 or 0.07 in these units is acceptable.

(e) [2 marks]

R1 The constant thermometer zero offset is systematic. Correct true temperatures are $1.5^{\circ}\mathrm{C}$ greater than the readings.

R1 Every point moves $1.5^{\circ}\mathrm{C}$ to the right, with its resistance unchanged. Temperature differences, and therefore the gradient, are unchanged.

(f) [1 mark]

R1 Stirring reduces temperature gradients in the water so the thermometer and wire are at approximately the same temperature. Accept an equivalent specific thermal-equilibration explanation.

1B Question 2 [10 marks]

(a) [1 mark]

R1 Writing $x = 1/E$ gives $d = Kx$: a constant gradient K and zero intercept. A nonzero intercept is not predicted by this model.

(b) [2 marks]

A1 $K = (33.6 - 12.6)/(1/3 - 1/8) = 100.8\,\text{MeV fm}$.

A1 $K_{\max} = (34.1 - 12.1)/(1/3 - 1/8) = 105.6$ and
$K_{\min} = (33.1 - 13.1)/(1/3 - 1/8) = 96.0\,\text{MeV fm}$, giving $K = (100.8 \pm 4.8)\,\text{MeV fm}$.

(c) [2 marks]

M1 $Z = K/(2 \times 1.44)$ and the same constant divisor applies to ΔZ.

A1 $Z = 35.0$ with $\Delta Z = 4.8/2.88 = 1.7$. Treating Z as physically integral does not remove the measurement uncertainty; the data are consistent with proton number 35.

(d) [2 marks]

M1 $R = 1.20(80)^{1/3} = 5.17\,\text{fm}$ and $E = K/R$.

A1 $E \approx 100.8/5.17 = 19.5\,\text{MeV}$. This is an estimate of the geometric scale at which the outside-point-charge model ceases to be reliable, not a guaranteed exact onset of a nuclear effect.

(e) [3 marks]

A1 The old model predicts $d = 100.8/12.0 = 8.40\,\text{fm}$, with $\Delta d = 4.8/12.0 = 0.40\,\text{fm}$.

R1 The predicted interval 8.0–8.8 fm does not overlap 5.5–6.5 fm, so the new result is inconsistent within the stated uncertainties.

R1 The disagreement alone does not establish a unique mechanism: an unmodelled systematic distance/energy bias, finite charge distribution or neglected target response must be checked. Accept another specific, scientifically justified model or calibration limitation; “human error” alone is insufficient.

End of worked solution book